AP PHYSICS 1

Unit 1 • Worksheet 1.4A • Answer key

Worksheet 1.4A • Reference Frames and Relative Motion

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Question 1

A train moves at 25 m/s. A passenger walks toward the front of the train at 2 m/s relative to the train. What is the passenger's speed relative to the ground?

(A) 23 m/s

(B) 27 m/s

(C) 25 m/s

(D) 2 m/s

Solution

Answer: (B). Same direction: $25 + 2 = 27\,\text{m/s}$.

Question 2

The same passenger now walks toward the back of the train at 2 m/s relative to the train. Her velocity relative to the ground is

(A) 23 m/s forward

(B) 27 m/s backward

(C) 23 m/s backward

(D) 2 m/s backward

Solution

Answer: (A). $+25 - 2 = +23\,\text{m/s}$: she still moves forward relative to the ground.

Question 3

Car A moves north at 30 m/s and car B moves north at 20 m/s. What is the velocity of B relative to A?

(A) 10 m/s north

(B) 50 m/s north

(C) 10 m/s south

(D) 50 m/s south

Solution

Answer: (C). $\vec{v}_{B/A} = \vec{v}_B - \vec{v}_A = 20 - 30 = -10\,\text{m/s}$: seen from A, car B moves backward (south).

Question 4

Two cars drive toward each other, each at 20 m/s relative to the road. What is the speed of one car relative to the other?

(A) 0
(B) 20 m/s
(C) 10 m/s
(D) 40 m/s

Solution
Answer: (D). Opposite directions: $20 - (-20) = 40$ m/s.

Question 5

A boat points straight across a river 60 m wide at 5 m/s relative to the water. The current is 2 m/s. How long does it take to cross?

(A) 12 s
(B) 30 s
(C) 8.6 s
(D) 11.1 s

Solution
Answer: (A). Only the velocity across the river matters: $60/5 = 12$ s.

Question 6

How far downstream does the boat in the previous question land?

(A) 12 m
(B) 24 m
(C) 60 m
(D) 30 m

Solution
Answer: (B). The current carries it $2 \text{ m/s} \times 12 \text{ s} = 24$ m downstream.

Question 7

A passenger in a train moving at constant velocity drops a ball. For the passenger, the path of the ball is

(A) straight down
(B) a curve bending backward
(C) a curve bending forward
(D) horizontal

Solution
Answer: (A). In the train's frame the ball starts at rest and only falls: straight down.

Question 8

For an observer standing beside the track, the path of the same ball is

(A) straight down
(B) a curve that moves forward with the train
(C) a curve that moves backward
(D) a horizontal line

Solution
Answer: (B). In the ground frame the ball also has the train's horizontal velocity: it follows a projectile path forward.

Question 9 (Free response)

An airplane heads north at 200 m/s relative to the air. A wind blows east at 50 m/s.

(a) Find the magnitude and direction of the plane's velocity relative to the ground.
(b) How far north does the plane travel in one hour?
(c) In which direction should the pilot point the plane to fly due north? Explain.

Solution
(a) $\sqrt{200^2 + 50^2} \approx 206\,\text{m/s}$, at $\tan^{-1}(50/200) \approx 14^\circ$ east of north. (b) $200 \times 3600 = 720\,000\,\text{m} = 720\,\text{km}$. (c) Slightly west of north, so that the westward component of the plane's velocity cancels the wind.